

Vemuri Shanmukha Chowdhary

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Education

Vellore Institute of Technology, Andhra Pradesh <i>Bachelor of Technology in CSE</i>	January 2022 - July 2026 CGPA: 8.22 / 10.00
Sri chaitanya Junior College, Vijayawada <i>Intermediate :Board Of Intermediate Education , Andhra Pradesh</i>	2020-2022 Percentage - 87.80%
Sri Chaitanya High School <i>Board of Secondary Education,Andhra Pradesh</i>	2019 - 2020 (CGPA: 10.0 / 10.00)

Technical Skills

- **Languages:** Python, Java
- **AI Agents & LLM:** Google ADK, Gemini API, MCP (Model Context Protocol), RAG, LLM Workflow Basics, Multi-Agent Systems, Prompt Engineering, Tool Calling
- **AI & ML:** Machine Learning, Deep Learning, NLP, Scikit-learn, OpenCV
- **Web & APIs:** Flask, FastAPI, REST APIs, Streamlit
- **Databases:** MySQL, MongoDB (Basics), SQLite, ChromaDB (Vector DB)
- **Developer Tools:** Git, GitHub, VS Code, Docker, CI/CD Basics

Work Experience

FlyRank AI <i>Machine Learning Engineering Intern</i>	Remote July 2026 – Present
• Contributing to machine learning projects involving data preprocessing, model development, and performance evaluation. • Applying Python and machine learning libraries to build and evaluate predictive models on real-world datasets.	

Projects and Research Experience

BizPilot AI – Autonomous Multi-Agent Business Operations Assistant [GitHub]	[Google ADK, MCP, RAG]
• Developing an autonomous business-operations assistant automating document search, email drafting, report generation, and business Q&A via AI agents, with a RAG pipeline (ChromaDB) for grounded document intelligence. • Architecting a Supervisor multi-agent system (Finance, HR, Document, SQL, Email, Calendar agents) coordinated through MCP servers, exposed via a Flask backend and Streamlit dashboard.	
Knowledge Based Search Engine [GitHub]	[Python, RAG]
• Developed an intelligent search engine that performs semantic search by converting both documents and user queries into high-dimensional vector embeddings. • Architected a Retrieval-Augmented Generation (RAG) system that retrieves the most relevant documents via vector search and then feeds that context to a Large Language Model (LLM) to generate a natural, accurate answer.	
Road Crash Severity Prediction Using Graph Neural Networks	[GNN, KNN]
• Developing a GNN-based crash severity prediction model, constructing a kNN-based graph from real-world crash datasets to capture spatial and relational dependencies between accident features. • Implemented and compared multiple Graph Neural Network models for crash severity prediction. — achieving a best accuracy with the Hybrid model, outperforming all baseline ML approaches.	

Achievements & Hackathons

- **Kaggle Community Hackathon** — Built and submitted an AI-powered business operations assistant.
- **INDIA RUNS – Data & AI Challenge** — Submitted an ML solution for the INDIA RUNS Data & AI Challenge.

Certifications

- **Oracle:** Oracle Gen AI Professional
- **MongoDB:** MongoDB Database Administrator
- **Microsoft:** Azure AI Fundamentals